UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 1

Date (or Date Range):02/02/2026 - 06/02/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Conducted feasibility study for online land allocation system in Botswana
- Documented current manual process (6 steps: physical form collection, postal submission, manual data entry, in-person queue checking)
- Selected tech stack: React, Node.js, Express, PostgreSQL
- Proposed AI chatbot integration for user support
- Prepared initial project proposal for supervisor meeting

3. Matters to Discuss with Supervisor

- Present feasibility assessment and tech stack for approval
- Discuss project scope (tribal and state land allocation only)
- Confirm problem statement accurately reflects real-world issues

4. Next Phase of the Project (as Agreed with Supervisor)

- Refine problem statement based on supervisor feedback
- Begin drafting detailed project proposal
- Research similar land management systems for reference

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5. References Consulted This Week

- Botswana Ministry of Land Management public service charter
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6. Additional Notes/Comments

(Any other relevant notes, reminders, or insights.)

- Supervisor approved project direction.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

- **Student Signature & Date: B.Ramokhua 08/02/2026**

- **Supervisor Signature & Date:**

Instructions

1. Complete this diary sheet weekly **before** meeting your supervisor.
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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 2

Date (or Date Range):09/02/2026 - 15/02/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Wrote detailed Problem Statement (6-step manual process documented)
- Wrote Project Motivation (personal experience: applied in 2021 at 17, still waiting)
- Defined Proposed Solution with AI chatbot feature
- Created system architecture diagram
- Defined functional requirements for 4 user roles

3. Matters to Discuss with Supervisor

- Review problem statement, motivation, proposed solution

4. Next Phase of the Project (as Agreed with Supervisor)

- Set up development environment
- Initialize backend with Express and Prisma
- Create database models for User, Application, LandBoard

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5. References Consulted This Week

4. Prisma ORM documentation

5. JWT authentication best practices

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6. Additional Notes/Comments

6. Problem statement approved. Moving to development phase.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

- **Student Signature & Date: B.Ramokhua 15/02/2026**

- **Supervisor Signature & Date:**

Instructions

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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 3

Date (or Date Range):16/02/2026 - 22/02/2026

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1. Student & Project Information

7. **Student Name: Boitsholo Ramokhua**
8. **Student ID: 202202152**
9. **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
10. **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Set up Node.js backend with Express
- Configured PostgreSQL database with Prisma ORM
- Created database schema: User, Application, LandBoard, WaitingListStat models
- Implemented JWT authentication (register/login)
- Created role-based middleware (Admin, Manager, Staff, Applicant)
- Implemented FIFO queue logic

3. Matters to Discuss with Supervisor

- Development environment demo and discussions

4. Next Phase of the Project (as Agreed with Supervisor)

- Begin frontend development with React and Vite
- Implement login/register pages
- Create role-based routing

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5. References Consulted This Week

- Node.js/Express documentation

- PostgreSQL documentation

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6. Additional Notes/Comments

- Ready for frontend.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

- **Student Signature & Date: B.Ramokhua 22/02/2026**

- **Supervisor Signature & Date:**

Instructions

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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 4

Date (or Date Range):23/02/2026 - 01/03/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Set up React frontend with Vite and Tailwind CSS
- Implemented authentication pages (Login, Register)
- Created AuthContext for global state management
- Implemented protected routes based on user roles
- Built Navbar component with role-based menu items
- Created Layout component for consistent page structure

3. Matters to Discuss with Supervisor

- (No formal meeting this week - work in progress)

4. Next Phase of the Project (as Agreed with Supervisor)

- Build multi-step application form
- Implement document upload component
- Create applicant dashboard

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5. References Consulted This Week

11. React Router v6 documentation

12. Tailwind CSS documentation

13. React Context API documentation

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6. Additional Notes/Comments

14. Frontend authentication flow complete. Role-based routing working.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

- **Student Signature & Date: B.Ramokhua 01/03/2026**

- **Supervisor Signature & Date:**

Instructions

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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 5

Date (or Date Range):02/03/2026 - 08/03/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Built multi-step application form (4 steps: Personal, Property, Documents, Review)
- Implemented document upload with Multer (24-hour temporary storage)
- Created API endpoints for application submission
- Seeded database with test data
- Set up real-time queue position polling (30-second intervals)

3. Matters to Discuss with Supervisor

- Development environment demo completed

4. Next Phase of the Project (as Agreed with Supervisor)

- Implement staff review dashboard
- Create manager analytics dashboard
- Build admin management panel

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5. References Consulted This Week

- Multer (file upload) documentation
- React Hook Form documentation

- Recharts documentation for analytics

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6. Additional Notes/Comments

15. Environment demo successful

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

- **Student Signature & Date: B.Ramokhua 08/03/2026**

- **Supervisor Signature & Date:**

Instructions

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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 6

Date (or Date Range):09/03/2026 - 15/03/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Built multi-step application form (4 steps: Personal, Property, Documents, Review)
- Implemented document upload with Multer (24-hour temporary storage)
- Created API endpoints for application submission
- Seeded database with test data
- Set up real-time queue position polling (30-second intervals)

3. Matters to Discuss with Supervisor(alpha feedback)

#	Feedback Point
1	Add gender allocation graph
2a	Lock Omang ID after submission
2b	Add asterisk for required fields
2c	Purpose field: text box → dropdown options
2d	Omang validation (digit 1=Male, 2=Female)
3	Document security at every layer
4a	AI as independent module/plugin
4b	Add link to AI feature (not embedded)

4c Focus on core first, AI later

4d Remove AI from formal objectives

5 Add current Land Board systems + citations to background

4. Next Phase of the Project (as Agreed with Supervisor)

- Implement staff review dashboard
- Create manager analytics dashboard
- Build admin management panel
- Implement gender allocation graph
- Fix application form validation and UI (asterisks, purpose dropdown, Omang lock)
- Document security layers
- Revise project objectives (remove AI as requirement)
- Research and cite current Land Board systems for background section
- Decouple AI into separate module/link

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5. References Consulted This Week

- Multer (file upload) documentation
- React Hook Form documentation
- Recharts documentation for analytics

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6. Additional Notes/Comments

- Major feedback: scope reduction. AI moved from core requirement to optional stretch goal.

  Objectives must be revised before alpha submission.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

16. **Student Signature & Date: B.Ramokhua 08/03/2026**

17. **Supervisor Signature & Date:**

Instructions

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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 7

Date (or Date Range):16/03/2026 - 22/03/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Added gender distribution donut chart to Manager Analytics
- Locked Omang number field after application submission
- Added red asterisk (*) to all required form fields
- Changed purpose field from text box to dropdown (Residential, Commercial, Agricultural, Industrial)
- Implemented Omang validation: middle digit 1=Male, 2=Female
- Documented security measures: JWT, RBAC, bcrypt, input validation, Prisma ORM
- Prepared tech stack defense documentation
- Decoupled AI from core system into separate /ai-assistant route
- Removed AI from formal project objectives (now a stretch goal)
- Added current Land Board systems + citations to report background section
- reviewed the LAKYC portal to see how it would fit in my project

3. Matters to Discuss with Supervisor(feedback)

- None this week (focused on implementation)

4. Next Phase of the Project (as Agreed with Supervisor)

- Prepare beta version for April 14 review

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5. References Consulted This Week

- Ministry land board process documentation
- OWASP security guidelines
- LAKYC government portal

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6. Additional Notes/Comments

- All alpha feedback completed. AI removed from objectives per supervisor advice.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

18. **Student Signature & Date: B.Ramokhua 22/03/2026**

19. **Supervisor Signature & Date:**

Instructions

1. Complete this diary sheet weekly **before** meeting your supervisor.
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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 8

Date (or Date Range):23/03/2026 - 29/03/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Fixed omangNumber schema issue (changed from Int to String for leading zeros)
- locked the omang number on the application form
- Created VerifyEmailPage.jsx to handle email verification links
- Updated email templates with correct frontend URL
- Added registration success modal with visual feedback
- Tested complete registration flow (signup → verify → login)
- Implemented document upload loading states

3. Matters to Discuss with Supervisor(feedback)

- no formal meeting

4. Next Phase of the Project (as Agreed with Supervisor)

- Start board-specific KPI implementation
- add a smart board matcher to help inform application decision making

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5. References Consulted This Week

- Prisma migration guides
- Nodemailer configuration

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6. Additional Notes/Comments

- Email verification 404 fixed. Registration flow now working end-to-end

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Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

20. **Student Signature & Date: B.Ramokhua 29/03/2026**

21. **Supervisor Signature & Date:**

Instructions

1. Complete this diary sheet weekly **before** meeting your supervisor.
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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 9

Date (or Date Range):30/03/2026 - 05/04/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Implemented document upload with 24-hour temporary storage expiry
- Created temporary file cleanup cron job
- Completed multi-step application form validation
- Implemented staff document review workflow (verify/reject)
- Completed in-app notification integration with email fallback
- Verified FAQ page with all 31 questions properly formatted

3. Matters to Discuss with Supervisor(feedback)

- no formal meeting

4. Next Phase of the Project (as Agreed with Supervisor)

- Complete remaining features and check(against the documented requirements) if i missed anything

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5. References Consulted This Week

- N/A

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6. Additional Notes/Comments

- All core features complete.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

- **Student Signature & Date: B.Ramokhua 05/04/2026**

- **Supervisor Signature & Date:**

Instructions

22. Complete this diary sheet weekly **before** meeting your supervisor.
23. Bring two copies: one to keep for your own records and one for your supervisor.
24. File your signed diary sheets safely; they will be submitted with your Progress and Final Reports.

UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 10

Date (or Date Range):06/04/2026 - 12/04/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Added PDF/Excel export functionality to analytics
- Implemented GPS location services for land board mapping
- Fixed registration bug
- Completed email verification flow testing
- Added registration success modal with navigation

3. Matters to Discuss with Supervisor(feedback)

- Document upload system approved
- Multi-step form structure approved
- Staff review workflow approved
- Move the audit trailing to manager dashboard
- fix the visualisations on the homepage to make them more meaningful

4. Next Phase of the Project (as Agreed with Supervisor)

- Complete remaining feedback (audit trail, board-specific KPIs)
- Make beta presentation slides

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5. References Consulted This Week

- N/A

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6. Additional Notes/Comments

- All core features complete.

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

25. **Student Signature & Date: B.Ramokhua 13/04/2026**

26. **Supervisor Signature & Date:**

Instructions

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UNIVERSITY OF BOTSWANA

DEPARTMENT OF COMPUTER SCIENCE

Weekly Diary Sheet

Academic Year: 2025 /2026

Week Number: Week 11

Date (or Date Range):13/04/2026 - 19/04/2026

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1. Student & Project Information

- **Student Name: Boitsholo Ramokhua**
- **Student ID: 202202152**
- **Project Title: KaboDitsha: A Web-Based Land Management System with AI Support for Botswana's Land Allocation Process**
- **Supervisor Name: Dr Lefoane**

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2. Work Done So Far

- Prepared beta presentation materials
- Fixed registration bug
- Completed email verification flow testing
- Added registration success modal with navigation

3. Matters to Discuss with Supervisor

- Beta presentation review

4. Next Phase of the Project (as Agreed with Supervisor)

- Prepare final presentation slides
- Finish final project report writing

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5. References Consulted This Week

- University of Botswana final project guidelines

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6. Additional Notes/Comments

- Beta version approved with pending improvements noted

Signatures

(Sign at the end of your meeting to confirm that both you and your supervisor have discussed these points.)

27. **Student Signature & Date: B.Ramokhua 20/04/2026**

28. **Supervisor Signature & Date:**

Instructions

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